Centrality

The main model:

1. DeBERTa (Zero-Shot Classifier): use it to categorize centrality and compute confidence https://huggingface.co/MoritzLaurer/deberta-v3-base-zeroshot-v2.0
    a. Trained by Microsoft that excels at reading a text and determining if a specific statement about that text is true.

Entire pipeline running on TDM studio process:

Note: TDM Studio does not connect to the network. Therefore, when applying models from Hugging Face, I downloaded the required files locally and loaded them from local storage.

1. Configuration
    a. Define all parameters and paths
    b. Model: DeBERTa
    c. Topic keyword: Abortion
    d. Date range: 2005-01-01 to 2025-12-31
    e. Sampling 100 articles/month
2. Helper Functions
    a. Extract Content
    b. Strip HTML
    c. Prepare text
    d. Contains topic
    e. Candidate labels
    f. Ordinal mapping
    g. Continuous sale based on ordinal probabilities
    h. Sampling 100 per month
3. Analyzer class
    a. Initialize DeBERTa zero shot classifier and output ordinal scale, continuous scale and confidence score
4. Load and validate Data
    a. Load the newspaper dataset
    b. Drop: Missing, corrupted entries and duplicates
5. Filter:
    a. Topic mentioned (articles that contains abortion)
    b. Date range (robust check to be in between 2005 to 2025)
    c. Monthly Sampling
6. Apply Centrality Model
    a. Run the model and store the output
7. Stored to result data frame
    a. Fields include:
        i. article_id
        ii. date
        iii. month
        iv. centrality_label
        v. confidence_score
8. Output to JSON in article level
9. Compute a summary table
10. Output the summary tables to EXCEL